DATA-DRIVEN APPROACH TO MINIMIZE DOWNTIME

FRESENIUS MEDICAL CARE

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Background

"Over 2 million people worldwide currently receive treatment with dialysis or a kidney transplant to stay alive, yet this number may only represent 10% of people who actually need treatment to live." - National Kidney Foundation

Fresenius Medical Care

Core Business: Renal Disease Solutions

Two fully automated production lines for Bibag

Bibag (Pouch used for cleaning hemodialysis equipment)

Problem:
Low Overall Equipment Effectiveness (OEE)

Breakdowns + → **Downtime** → - **OEE**

Problem Statement

The project aims to develop an **integrated reporting system** for the **failure modes** of all workstations, with the objective of **detecting bottlenecks** and **reducing the downtime** in station 280 by up to 70% within 4 months. The project scope includes implementing a **standardized database** and **Standard Operating Procedures (SOPs)**.

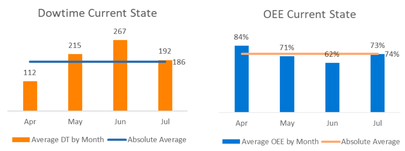
Failure Mode	% of Total Downtime	Success Rate	
		100.00%	70.00%
Touchtest	52.10%	52.10%	36.47%
Vacuum Issues	15.13%	15.13%	10.59%
Sealed Empty Bags	2.52%	2.52%	1.76%
Bag Opening Issues	1.68%	1.68%	1.18%
Bag position	0.84%	0.84%	0.59%
Grand Total	72.27%	72.27%	50.59%

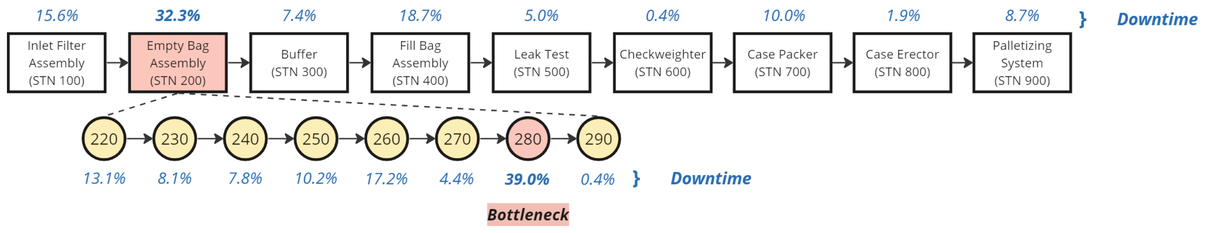
- Failure detection process
- Revenue
- Motivation
- Employee engagement
- Strategic decision-making

Current State

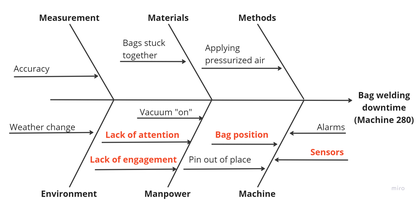
There is low variation on the process, no alterations on the line and the problems can be predicted. Breakdowns were the main disruption analyzed.

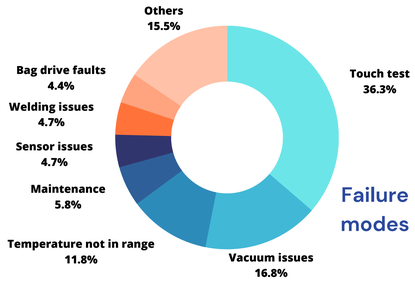
***Downtime = 186 min/day** OEE = 74%**

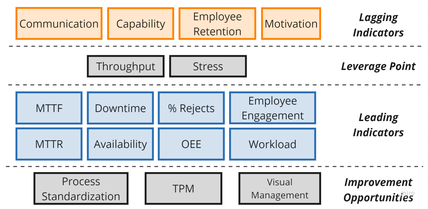




The data collection process is not standardized, and downtime is estimated by the operator. Besides that, there is no communication between shifts, and most problems are not registered.







*4 months data

All the data was provided by the company.

Future State

SOP for station 280

Minimize breakdowns and downtime for station 280.

Data Entry SOP

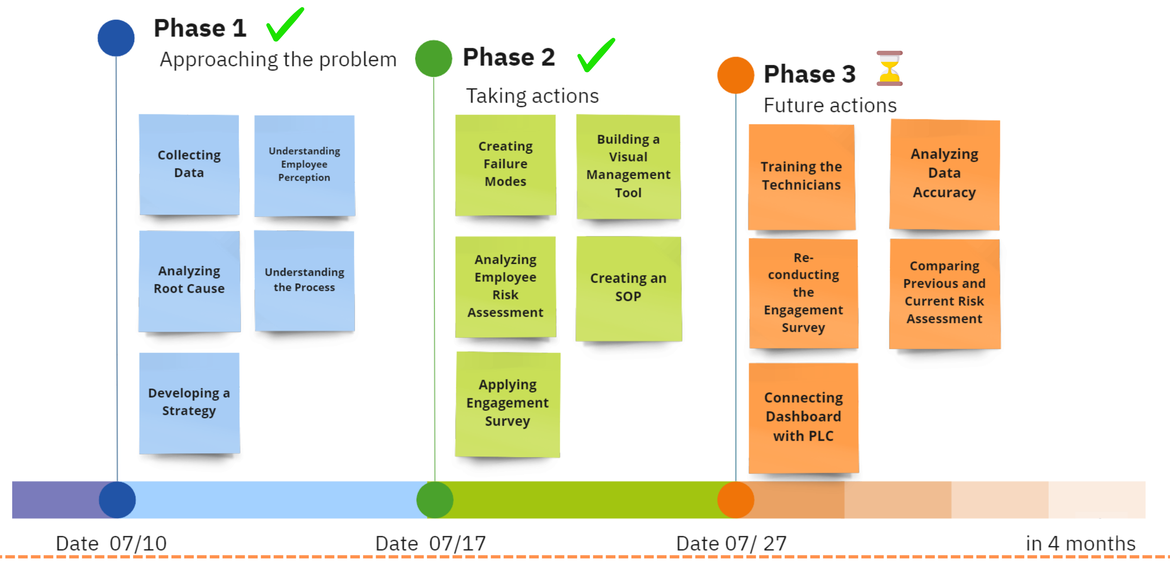
Standardize the data entry to generate more accurate reports for bibag line.

- Increase data accuracy
- Improve Communication
- Enhance visibility
- Reduce Workload

Indicators	Actual	50%	%	70%	%
Performance	98.75%	98.75%	-	98.75%	-
Quality	99.60%	99.60%	-	99.60%	-
Availability	**75.46%**	**76.61%**	1.14 p.p	**77.06%**	1.60 p.p
OEE	**74.22%**	**75.35%**	**1.13 p.p**	**75.80%**	**1.58 p.p**
Products / Min	33.33	33.33	-	33.33	-
Available Min / Shift	**543.32**	**551.56**	1.52 %	**554.85**	2.12 %
Total Bags Produced / Day	**35,769**	**36,311**	1.52 %	**36,528**	2.12 %
Estimated increase in bag production / Day	-	**542**	-	**759**	-
Estimated Price / Bag*	$ 6.93	$ 6.93	-	$ 6.93	-
REVENUE INCREASE / YEAR	-	***$ 1.3M***	-	***$ 1.9M***	-

*Considering 2.5 distributor markup | **353 days

Action Plan



Sustainability

Risk Assessment	
Category	**Risk**
Motivation	3.63
Engagement	3.25
Cultural alignment	3.33

LOW	MEDIUM	HIGH
4-5	3-3.99	1-2.99

CATEGORIES	SUB-CATEGORIES	AVERAGE	RISK
Motivation	Employee Needs	3.50	MEDIUM
	Workload	4.25	LOW
	Meaningful work	3.92	MEDIUM
	Work-related affect (positive and negative)	***3.19***	***MEDIUM***
	Overall workplace safety	3.50	MEDIUM
	Communication	***3.44***	***MEDIUM***
Engagement	***Satisfaction***	***3.00***	***MEDIUM***
	Productivity	***3.31***	***MEDIUM***
Cultural Alignment	Supportive work culture	3.54	MEDIUM
	Problems at work	***3.00***	***MEDIUM***